



2025 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

DO NOT REMOVE PAPER FROM EXAMINATION ROOM

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Centre Number

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Student Number

Physics

Morning Session
Friday, 8 August 2025

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- Use the Multiple-Choice Answer Sheet provided
- A data sheet, formulae sheet and Periodic Table are provided SEPARATELY
- Write your Centre Number and Student Number on the top of this page

Total marks:
100

Section I – 20 marks (pages 2–11)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 12–36)

- Attempt Questions 21–35
- Allow about 2 hours and 25 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the Multiple-Choice Answer Sheet for Questions 1–20.

- 1 A current-carrying conductor is placed within a magnetic field.

The conductor will experience no force when

- A. the current in the conductor is increased.
- B. the current in the conductor is kept constant.
- C. the conductor is parallel to the magnetic field.
- D. the conductor is perpendicular to the magnetic field.

- 2 In a thought experiment a laser pointer is shone out the front of a jet travelling at $0.4c$ relative to Earth.

What is the speed of light as measured by a stationary observer, also relative to Earth?

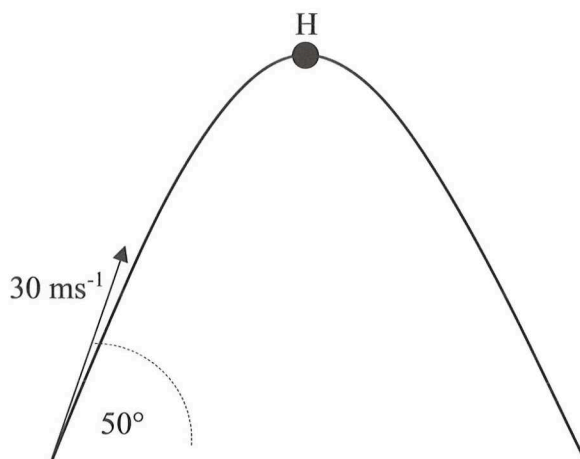
- A. $0.1c$
- B. $0.4c$
- C. $1.0c$
- D. $1.4c$

- 3 Which of the following data is used to determine the density of a star?

- A. Doppler shift
- B. Peak wavelength
- C. Absorption spectrum
- D. Width of spectral lines

- 4 A projectile is launched from the ground at an angle of 50° to the horizontal with a velocity of 30 ms^{-1} , as shown in the diagram.

H is the highest point of its motion.



What is the height of H above the ground?

- A. 1.2 m
 - B. 19.0 m
 - C. 26.9 m
 - D. 42.2 m
- 5 In a demonstration modelling nuclear fission, a teacher placed ten loaded mousetraps, each loaded with a ball, on the floor.

The teacher then dropped another ball onto the mousetraps. In the moments after the ball was dropped, the students observed that all the mousetraps released their balls.

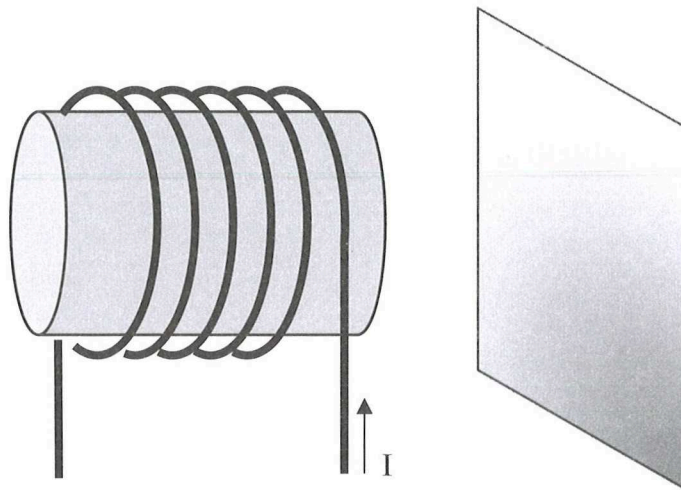
What do the balls represent in this demonstration?

- A. Atoms
- B. Electrons
- C. Neutrons
- D. Protons

- 6 An AC induction motor relies on several pairs of stator coils, set in a circle, producing a rotating magnetic field. This relative motion of the magnetic field induces an emf in the rotor.

Which of the following statements describes the relative motion of the rotor and the external rotating magnetic field provided by the stator coils in an AC induction motor?

- A. The rotor rotates at the same speed as the external magnetic field.
 - B. The rotor rotates slower than the external magnetic field.
 - C. The rotor rotates faster than the external magnetic field.
 - D. The rotor remains stationary.
- 7 A horizontal solenoid, comprised of a coil of wire around a solid core, is placed near a conductive metal plate, as shown in the diagram. The current in the solenoid is suddenly increased, strengthening the magnetic field around it.



Which of the following statements correctly describes what happens in the metal plate as the current in the solenoid increases?

- A. No induced currents are generated because the metal plate is not connected to a circuit.
- B. A temporary induced eddy current forms, resulting in a magnetic north pole facing the solenoid.
- C. The solenoid induces a constant eddy current, which continues even after the solenoid current stops changing.
- D. A temporary induced eddy current forms, generating a magnetic field that opposes the increasing magnetic field of the solenoid.

- 8 Maxwell made predictions about the production of electromagnetic (EM) waves. He concluded they were made of oscillating, perpendicular fields.

Which of the following statements about the production of the fields is correct?

- A. Maxwell predicted that accelerating particles create a changing electric field that creates a changing magnetic field.
 - B. Maxwell predicted that stationary charged particles create a changing magnetic field that creates a changing electric field.
 - C. Maxwell predicted that accelerating charged particles create a changing gravitational field that creates a changing magnetic field.
 - D. Maxwell predicted that stationary charged particles create a changing electric field that creates a changing gravitational field.
- 9 Two identical polarised sunglasses are placed in front of an unpolarised light source as shown.



Which of the following correctly shows what will happen if the sunglasses are rotated relative to each other?

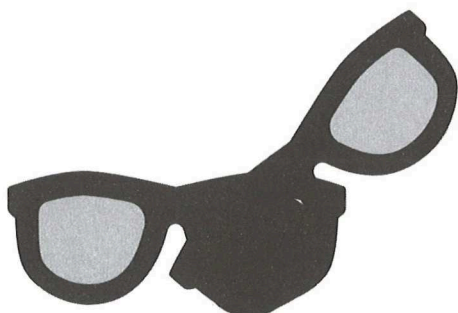
A.



B.



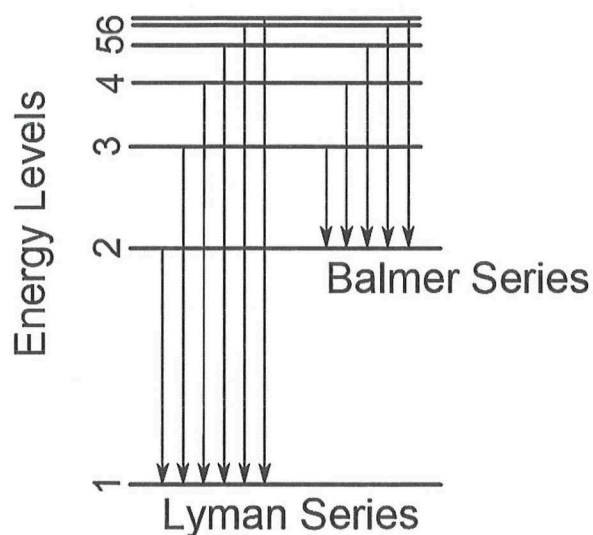
C.



D.



- 10 The hydrogen spectrum produces light in two series as shown below.



A spectral line in the Balmer Series of hydrogen has a wavelength of 397 nm.

Which transition in energy levels results in this wavelength?

- A. Level 5 to 2
 - B. Level 5 to 1
 - C. Level 6 to 2
 - D. Level 7 to 2
- 11 The orbital period of Earth is 365.25 days and the average distance between the Earth and the Sun is 1.496×10^{11} m.

What is the mass of the Sun?

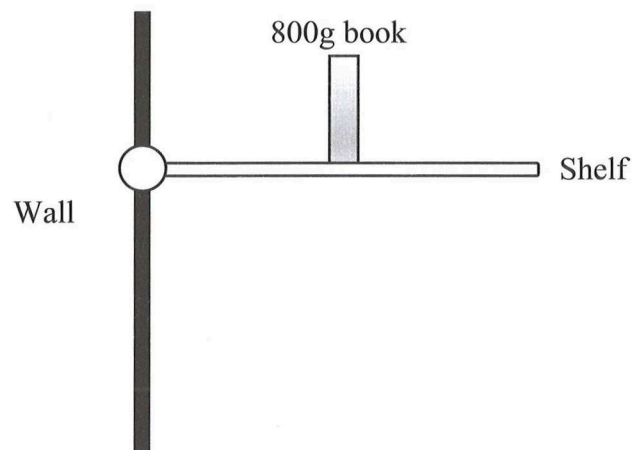
- A. 1.33×10^{19} kg
- B. 1.99×10^{30} kg
- C. 1.15×10^{33} kg
- D. 7.16×10^{33} kg

- 12 Satellite A has a mass of m and an orbital radius of $4r$, while satellite B with a mass $4m$ and an orbital radius of r . Both satellites orbit the same object.

Which of the following statements is true?

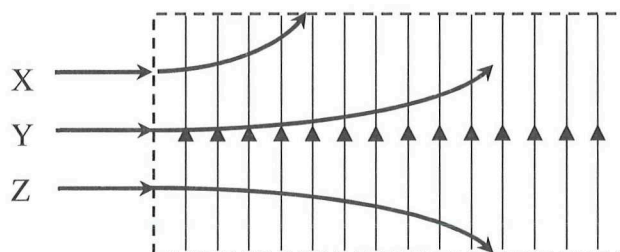
- A. They have the same orbital velocity.
 - B. Orbital velocity of satellite A is twice that of satellite B.
 - C. Orbital velocity of satellite B is twice that of satellite A.
 - D. Orbital velocity of satellite B is 4 times that of satellite A.
- 13 A massless shelf with a length of 1 m, is hinged to a wall. The shelf supports an 800 g book halfway along its length.

What is the torque in Nm produced by the weight of the book?



- A. 3.92
- B. 7.84
- C. 3920
- D. 7840

- 14 The brightest star in the Orion constellation, Rigel, has a power output of 4.35×10^{31} W. How much of Rigel's mass is being transformed into energy, each second, to produce this power output?
- A. 1.37×10^{14} kg
 B. 4.83×10^{14} kg
 C. 1.37×10^{15} kg
 D. 1.45×10^{23} kg
- 15 Which of the following postulates correctly identifies Schrödinger's contribution to our understanding of particle physics?
- A. Electrons have a wave nature and exist in orbitals.
 B. Electrons orbit the atomic nucleus in discrete energy levels.
 C. A small, dense positively charged nucleus is surrounded by orbiting electrons.
 D. Electrons have a dual wave-particle nature existing in stable standing waves around the nucleus.
- 16 Three charged particles, X, Y and Z, pass through a uniform electric field as shown in the diagram.

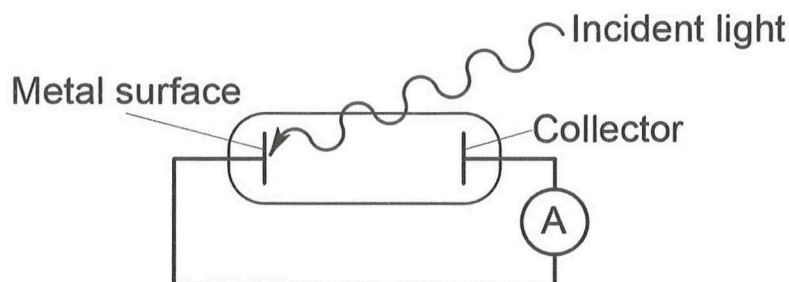


Assume that the masses are identical and that the initial velocities are the same.

Which of the following correctly identifies the relative nature of the three charges?

	X	Y	Z
A.	-2q	-q	+q
B.	-q	-2q	+2q
C.	+q	+2q	-2q
D.	+2q	+q	-q

- 17 The apparatus pictured below can be used to investigate the photoelectric effect and gather evidence about whether light is composed of particles or waves.



Some observations made are

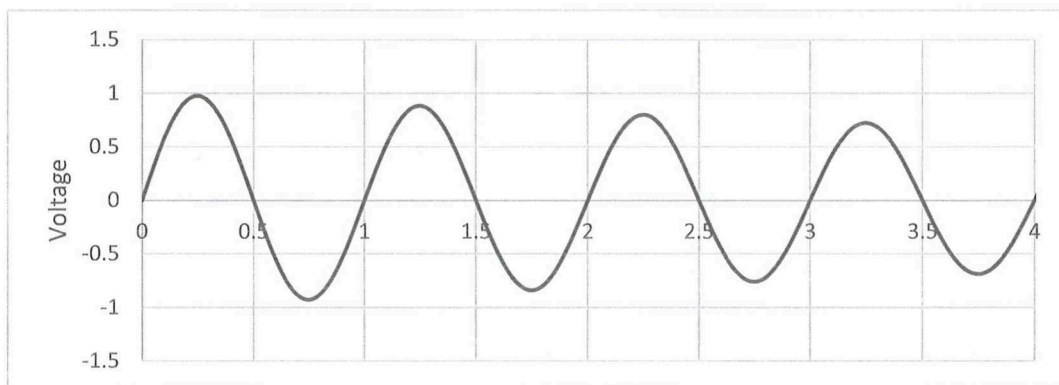
1. electrons are emitted from metal surfaces upon exposure to incident light
2. electrons are only emitted above a certain threshold frequency of incident light
3. electrons are emitted immediately when incident light is above the threshold frequency regardless of light intensity
4. at frequencies above the threshold frequency, the number of electrons emitted are determined only by the intensity of light.

Which observation above could be consistent with the wave model of light?

- A. 1
- B. 2
- C. 3
- D. 4

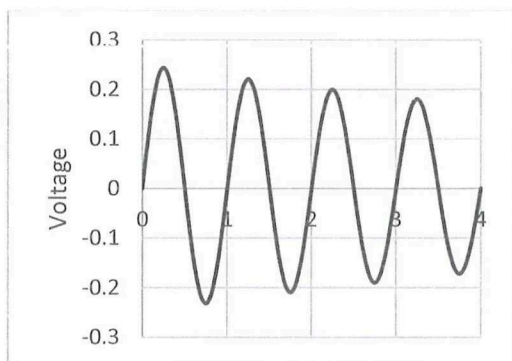
- 18 A transformer is set up so that the primary coil has 400 turns and the secondary coil has 100 turns.

The input voltage to the primary coil is shown on the graph below:

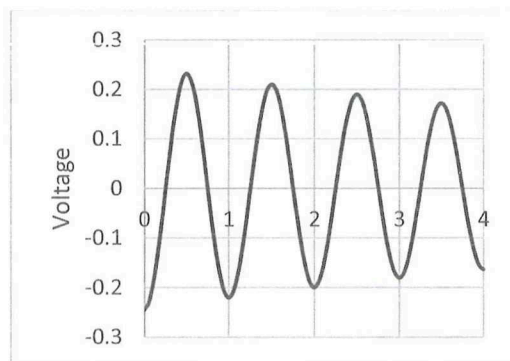


Which of the following graphs shows the output voltage for this transformer?

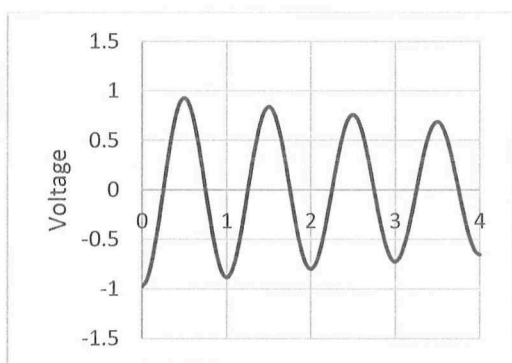
A.



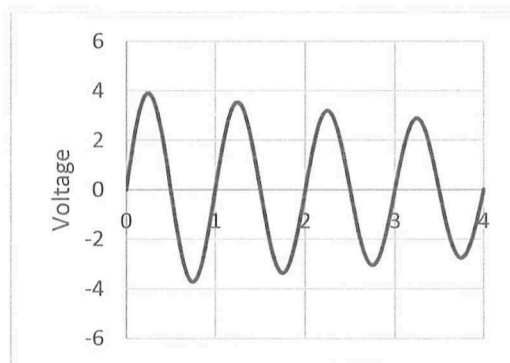
B.



C.



D.

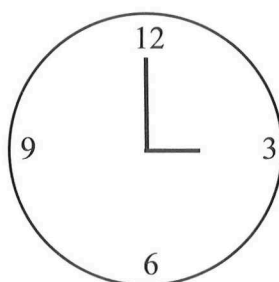


- 19 A 100 kg object is projected from a point located 6371 km from the centre of the Earth.

What is the launch velocity of the projectile so that the velocity would never fall below 2000 ms^{-1} ?

- A. 5604 ms^{-1}
- B. 11208 ms^{-1}
- C. 11386 ms^{-1}
- D. 13209 ms^{-1}

- 20 A student was observing the motion of the minute hand on a clock with a length of 5 cm. They noticed that it takes 15 minutes for the minute hand to move from 12 to 3.



Which row of the table shows the correct angular velocity (rad/s) and average speed (m/s) of the minute hand at a point furthest from the centre of the clock?

	<i>Angular velocity (rad/s)</i>	<i>Average speed (m/s)</i>
A.	105×10^{-3}	7.25×10^{-3}
B.	21.0×10^{-3}	7.25×10^{-5}
C.	1.75×10^{-3}	8.73×10^{-5}
D.	17.5×10^{-3}	8.73×10^{-3}

Section II

80 marks

Attempt Questions 21–35

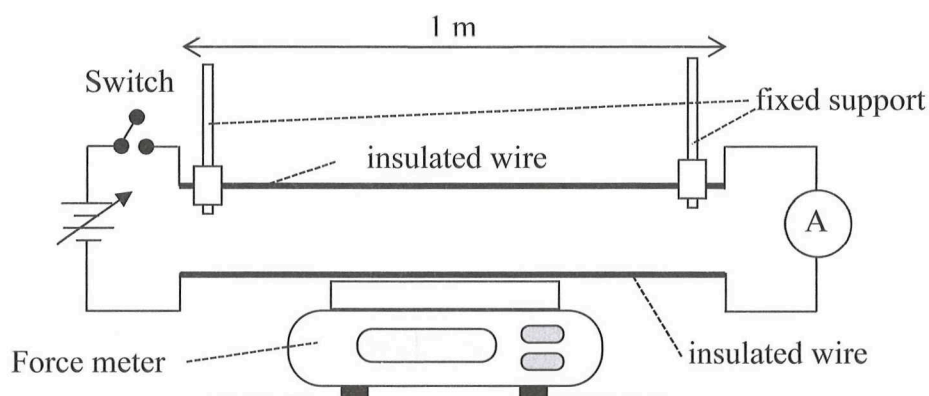
Allow about 2 hours and 25 minutes for this section

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- EXTRA writing booklets are available if required. If you use an EXTRA writing booklet, clearly indicate which question you are answering by writing the question number before beginning the response.

Question 21 (6 marks)

Students conducted an experiment to investigate the relationship between the current in parallel wires and the force experienced between them.

Their apparatus is shown below:



Before the investigation starts, the students set the force meter to read 0.0000 N.

- (a) What is the direction of the magnetic field in the region between the insulated wires after the switch is closed? 1

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Question 21 continues on page 13

Question 21 (continued)

- (b) When the ammeter displayed a current of 14.4 A, the force meter display was 0.0138 N. **2**
Determine the distance between the two wires.

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- (c) Explain why the reading of the force meter increased after the switch was closed. **3**

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End of Question 21

Question 22 (4 marks)

A space agency is planning to place a communications satellite in geostationary orbit.

- (a) Use Kepler's Laws to calculate the altitude of geostationary orbit from the Earth's surface. **2**

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- (b) Justify the need to place a communications satellite in a geostationary orbit rather than a low Earth orbit. **2**

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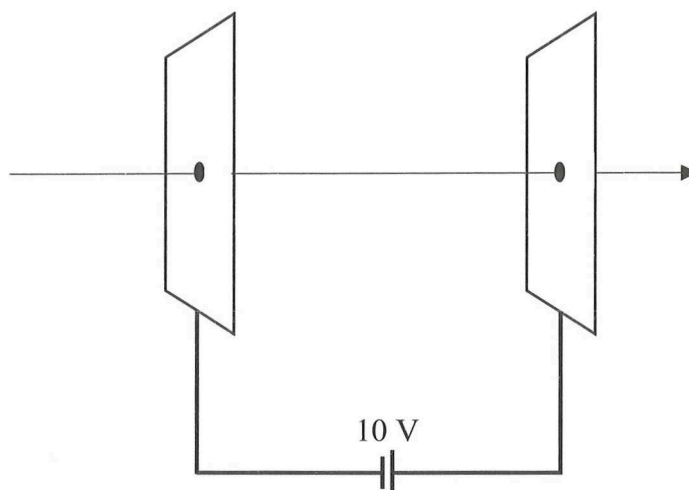
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Question 23 (4 marks)

An electron is accelerated using an electric field created between two charged plates. The electron enters through a small opening and exits through an opening on the oppositely charged plate as shown by the arrow. The electric field only turns on once the electron has entered the field. The plates are separated by 0.04 m and have a potential difference of 10 V.



- (a) Calculate the electric field strength between the plates. 1

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- (b) If the electron enters the field at $3.6 \times 10^5 \text{ ms}^{-1}$, calculate its velocity as it exits the field. 3

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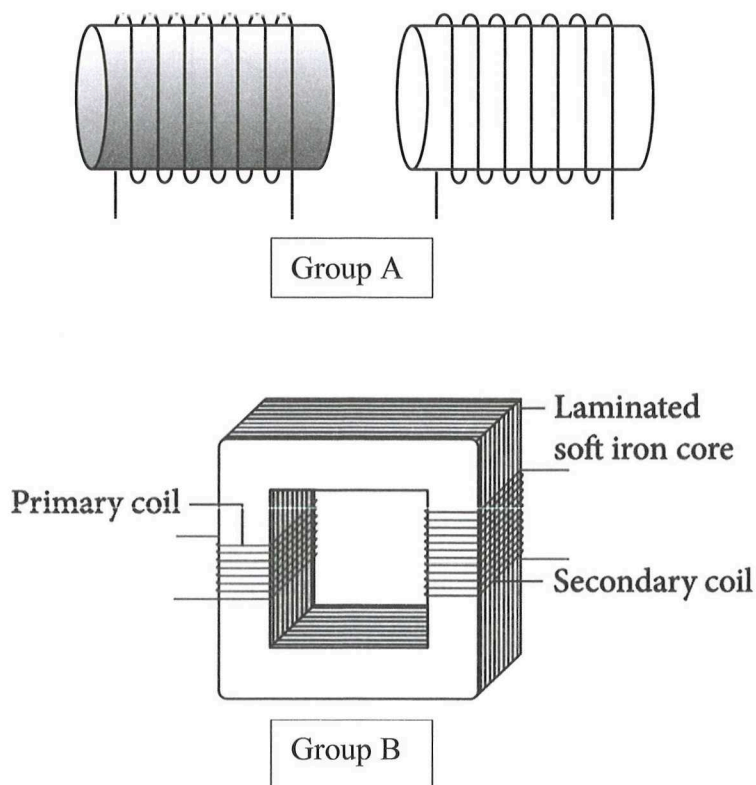
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Question 24 (8 marks)

Two groups of students construct transformers. Group A builds a simple transformer by placing two coils adjacent to each other, one coil around a solid iron core and the other around a hollow paper tube, as shown. Group B wraps the coils around a laminated iron core as shown.



- (a) Explain why the transformer built by Group B showed greater efficiency than the transformer built by Group A.

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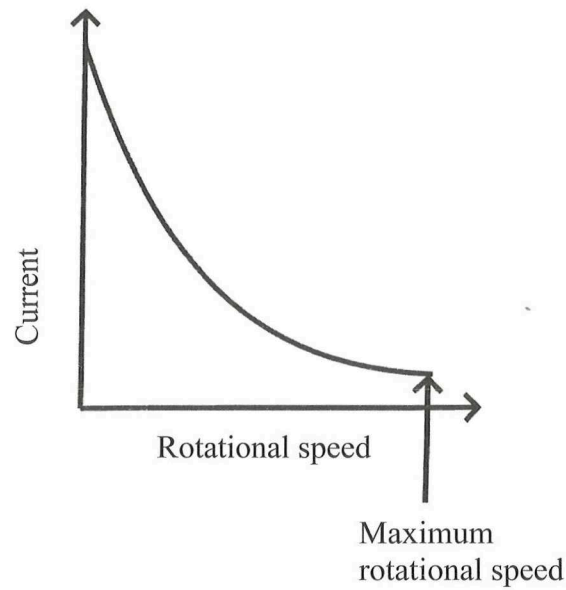
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Question 24 continues on page 17

Question 24 (continued)

- (b) A student performs experiments with a simple DC motor. The rotational speed and current in the DC motor were recorded below: 4



Explain why the current decreases as rotational speed increases with reference to the conservation of energy.

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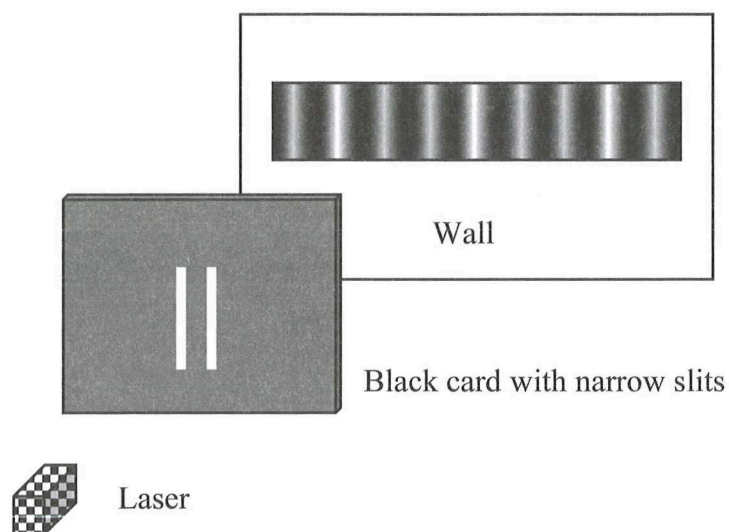
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End of Question 24

Question 25 (4 marks)

Light from a laser is incident on a black card with two narrow slits cut into it. After the light passes through, the following pattern is observed on a white wall. **4**



Using an equation, explain any alterations that could be made to the apparatus to increase the distance between bright and dark bands on the wall, assuming the laser wavelength cannot be changed.

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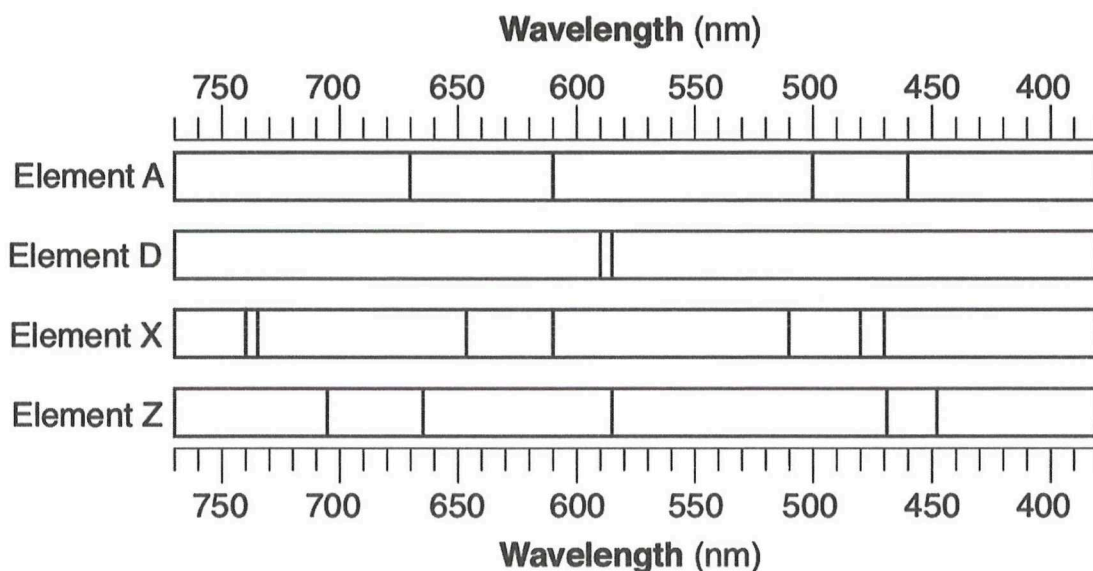
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Question 26 is on page 20

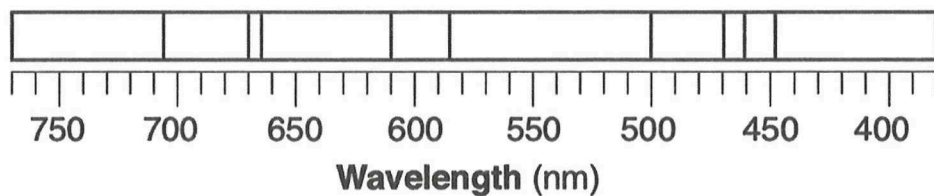
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Question 26 (5 marks)

Below are the absorption spectra of four known elements on Earth:



Below is an absorption spectrum of a new, nearby star.



- (a) Use the information above to identify the two elements present in the atmosphere of the star. 2

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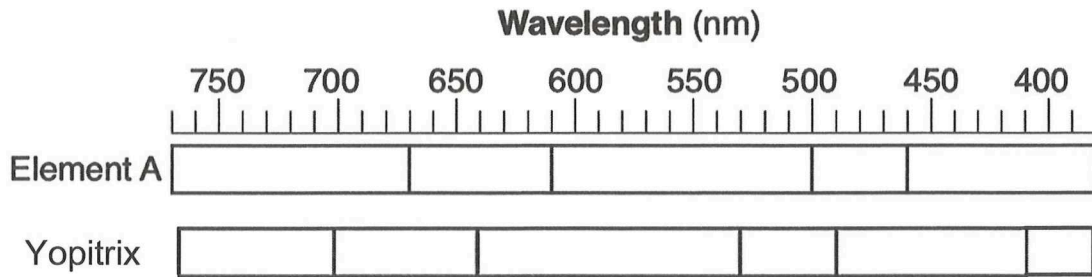
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Question 26 continues on page 21

Question 26 (continued)

- (b) The absorption spectrum below, labelled Yopitrix, was gathered from a distant star.

3



The distant star is called Yopitrix and the star is known to have Element A in its atmosphere.

Use the stimulus above to determine if Yopitrix is moving away from, or towards, Earth and explain how its velocity, relative to Earth, could be determined.

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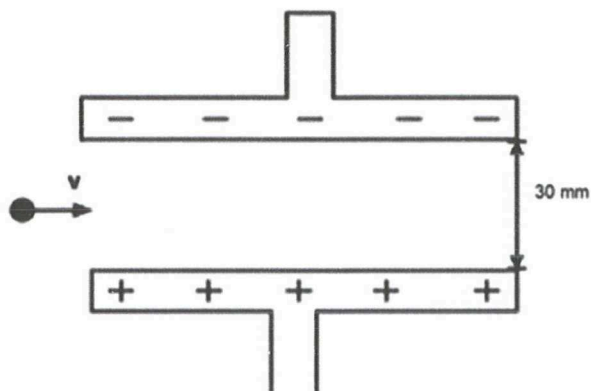
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End of Question 26

Question 27 (6 marks)

A proton passes into the space between oppositely charged, parallel plates, initially moving parallel to them. The plates are separated by a distance of 30 mm. The effect of gravity on its motion can be ignored.



The table below shows vertical displacements of the proton measured at different times.

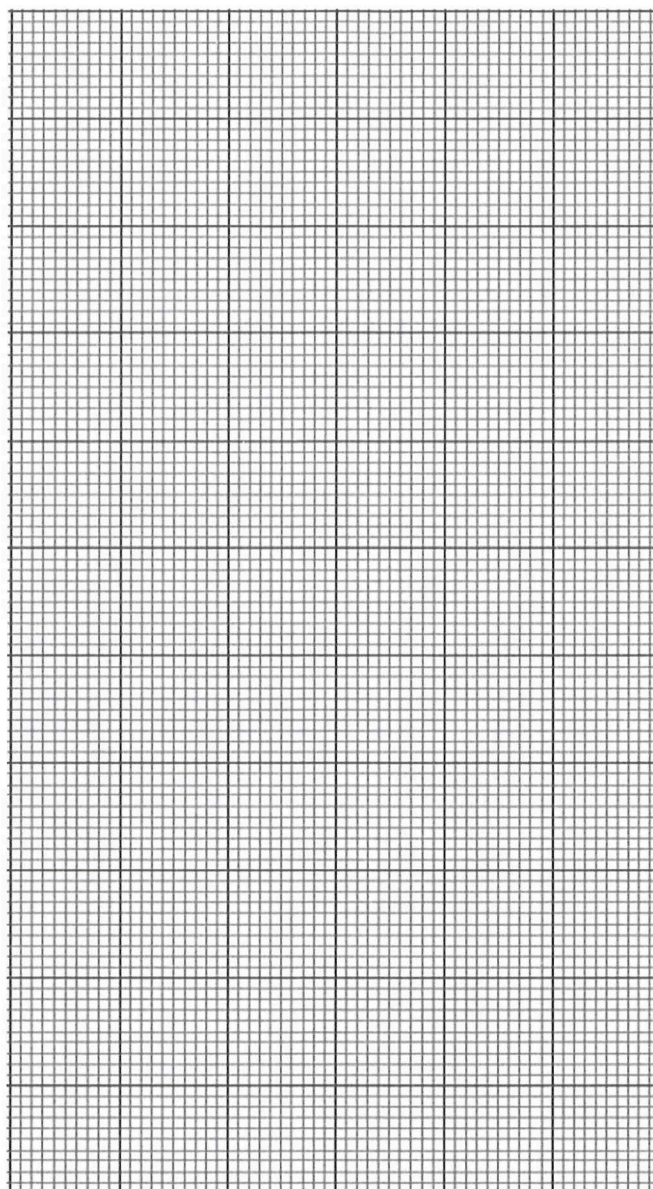
<i>Time</i> ($\times 10^{-9}$ s)	<i>Time</i> ² ($\times 10^{-18}$ s ²)	<i>Vertical displacement up</i> ($\times 10^{-8}$ m)
0	0	0
1	1	2
2	4	8
3	9	17
4	16	30
5	25	48

Question 27 continues on page 23

Question 27 (continued)

- (a) Draw a graph of vertical displacement of the particle versus time squared AND use your graph to determine the approximate acceleration. Show all working.

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Question 27 continues on page 24

Question 27 (continued)

- (b) Determine the time the proton takes to hit the top plate from the moment it enters the electric field. 2

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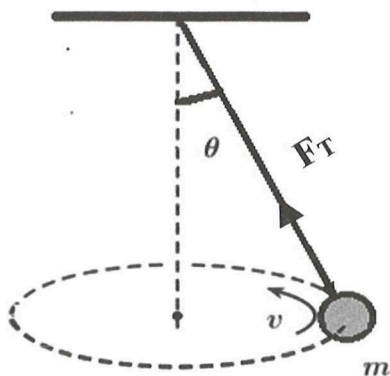
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End of Question 27

Question 28 (5 marks)

A bob with a mass of 0.5 kg attached to a string is connected to a DC motor that rotates at an angular velocity of 20 rad/s. As a result, the bob undergoes uniform circular motion with a horizontal radius of 0.3 m.



- (a) Calculate the linear speed of the bob.

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- (b) Use a clearly labelled vector diagram of the forces acting to determine the angle that the string makes with the vertical.

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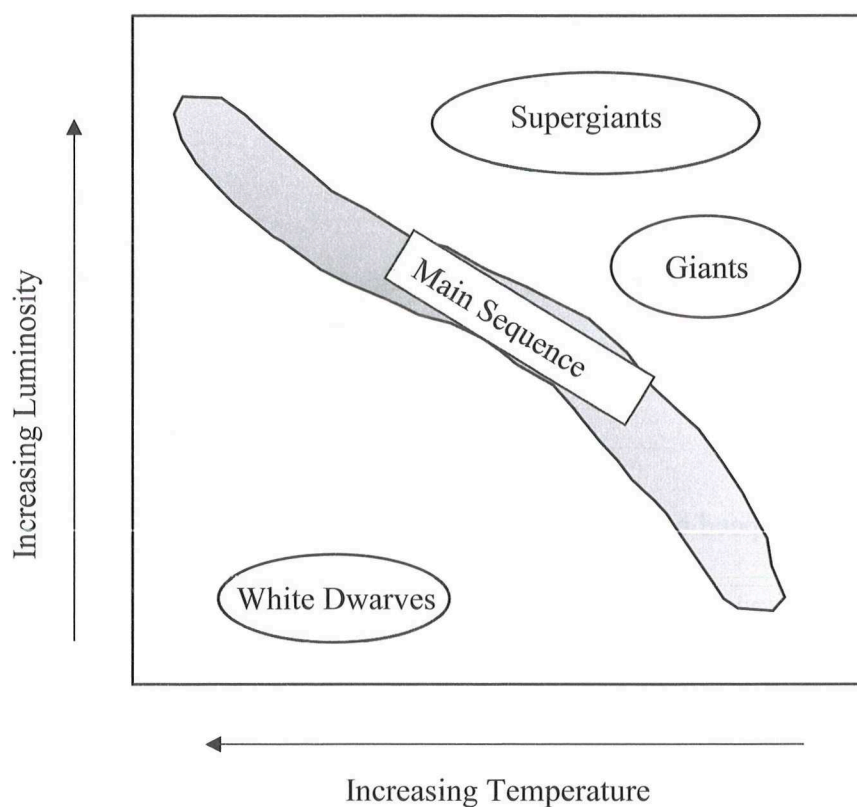
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Question 29 (4 marks)

Below is a simplified Hertzsprung-Russel diagram used to classify stars based on their temperature and luminosity. The major classifications of stars are found within the three ellipses and the main sequence stars are identified.



- (a) On the diagram above, identify with an X on the main sequence where a White Dwarf may have begun its evolution. 1

Question 29 continues on page 27

Question 29 (continued)

- (b) Explain how the evolution of a Supergiant is different to that of a White Dwarf.

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In your answer, refer to the types of nucleosynthesis reactions occurring throughout their lifetimes.

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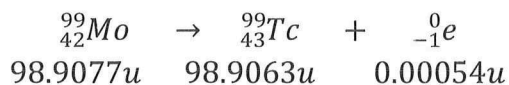
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End of Question 29

Question 30 (5 marks)

The natural radioactive decay of molybdenum-99 produces technetium-99, a radioisotope used in nuclear medicine. The nuclear reaction equation is shown with the mass of each species in atomic mass units.



- (a) Show that the energy released by one decay is 0.8011 MeV.

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- (b) The half-life of molybdenum-99 is 65.94 hours.

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Calculate the percentage of atoms of molybdenum-99 that have decayed in a 5-hour period.

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Question 31 (4 marks)

- (a) Calculate the momentum of an electron traveling at $0.4c$.

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- (b) Explain why the momentum of an electron traveling at $0.8c$ is more than double that of an electron traveling at $0.4c$.

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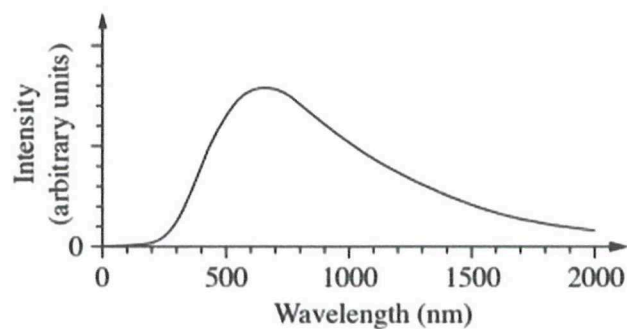
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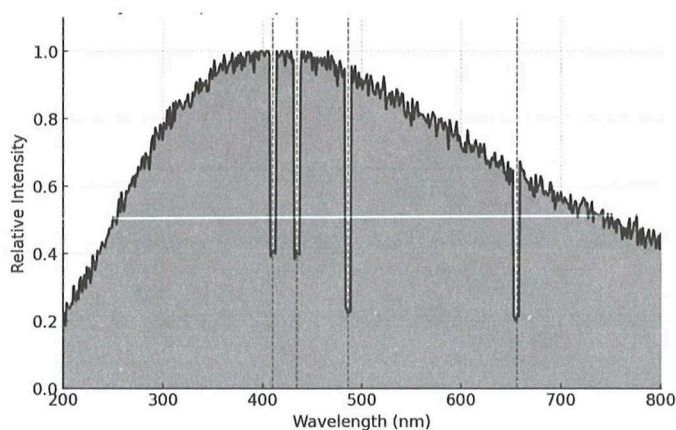
Question 32 (5 marks)

Two electromagnetic radiation curves are shown. Curve A is for a hot body on Earth. Curve B is the radiation from a star.

Curve A



Curve B



- (a) Account for the similarities and differences of the two curves.

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- (b) Calculate the surface temperature of the star.

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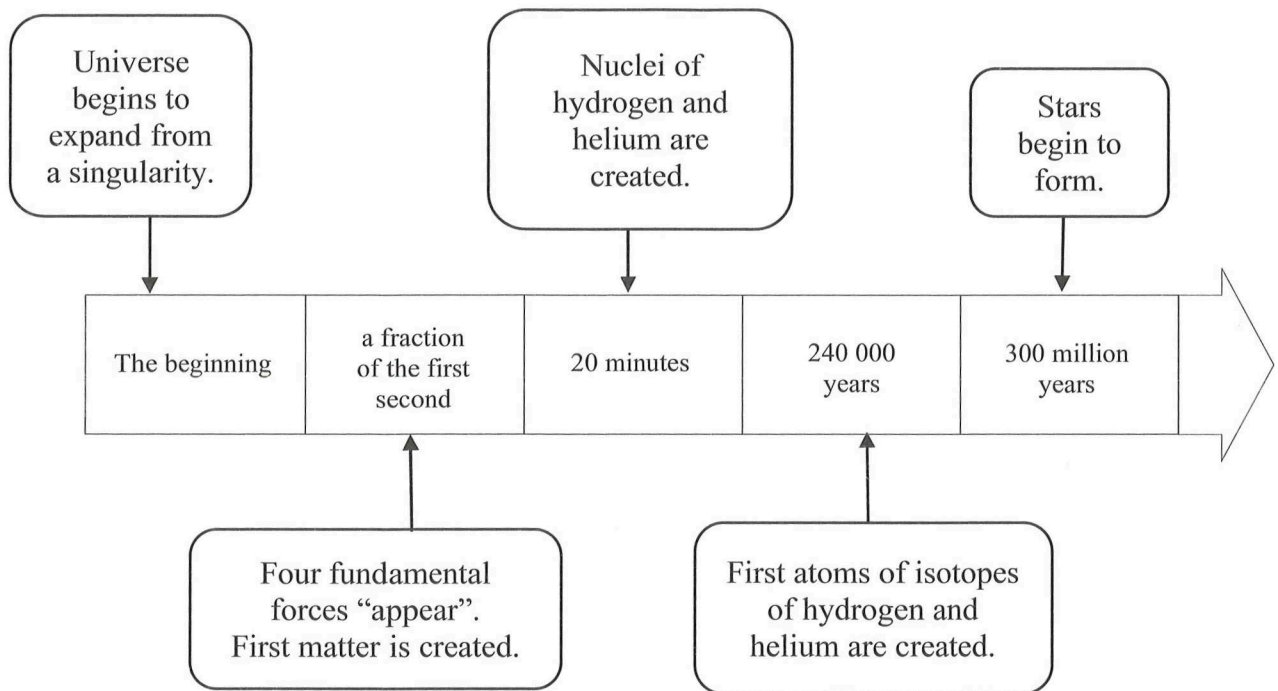
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Question 33 (4 marks)

Below is a simplified timeline of the formation of forces and matter after the Big Bang. All times are approximate.

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Using the information above, outline the creation of matter and the role of forces in the formation of the first atoms of helium-4.

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Question 34 (8 marks)

The table below shows the properties of quarks and hadrons. Particle accelerators, like those used in high-energy physics experiments, are designed to accelerate protons and other particles to extremely high speeds. These particles then collide with each other, allowing scientists to study their interactions and the fundamental forces of nature. In some cases, the mass and energies involved in the proton collisions are converted into new fundamental particles.

8

<i>Particle</i>	<i>Quark composition</i>	<i>Charge</i>	<i>Rest mass (approx.)</i>
Up Quark (u)	-	$+\frac{2}{3}e$	2.2 MeV/c ²
Down Quark (d)	-	$-\frac{1}{3}e$	4.7 MeV/c ²
Charm Quark (c)	-	$+\frac{2}{3}e$	1.27 GeV/c ²
Strange Quark (s)	-	$-\frac{1}{3}e$	96 MeV/c ²
Top Quark (t)	-	$+\frac{2}{3}e$	173 GeV/c ²
Bottom Quark (b)	-	$-\frac{1}{3}e$	4.18 GeV/c ²
Proton	uud	+ 1	938 MeV/c ²
Neutron	udd	0	939 MeV/c ²

Using the properties of quarks in the table above, explain how the collision of protons in a particle accelerator could lead to the production of heavier quarks, such as the top and bottom quarks. You may ignore any quark-gluon interactions.

In your answer, refer to

- the operation of particle accelerators and show calculations for energy–mass equivalence of the top and bottom quarks and two protons
- relativistic momentum of the colliding protons
- velocity of protons required to produce one top and one bottom quark in a single collision, if the quarks are considered at rest after the collision.

Question 34 continues on page 33

Question 34 (continued)

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Question 34 continues on page 34

Question 34 (continued)

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End of Question 34

Question 35 (8 marks)

Throughout history, experimental evidence has played a crucial role in developing new models, validating existing models, and abandoning previously accepted models in physics.

8

Justify this statement by referring to specific experiments and their evidence.

In your response, provide examples of how experimental results have led to the creation of new models, confirmed established models, and caused outdated models to be discarded, shaping our understanding of physics.

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Question 35 continues on page 36

Question 35 (continued)

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End of Examination

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Section II extra writing space

If you use this space, clearly indicate which question you are answering by writing the question number before beginning the response.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Section II extra writing space

If you use this space, clearly indicate which question you are answering by writing the question number before beginning the response.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering by writing the question number before beginning the response.

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EXAMINERS

Andrew Latham (Convenor)
Lily Okati
Janet Pemberton
Peter Robson
Adam Sloan

Stella Maris College, Manly
Santa Sabina College, Strathfield
Roseville College, Roseville
William Clarke College, Kellyville
Knox Grammar School, Wahroonga

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2025 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

MARKING GUIDELINES

Physics

Section I

20 marks

Questions 1–20 (1 mark each)

Question	Answer	Outcomes Assessed	Targeted Performance Band
1	C	PH 12-13	2–3
2	C	PH 12-14	2–3
3	D	PH 12-14	2–4
4	C	PH 12-12, PH 12-4	2–4
5	C	PH 12-15, PH 12-2	2–4
6	B	PH 12-13	2–4
7	D	PH 12-13, PH 12-5	2–4
8	A	PH 12-14	2–4
9	A	PH 12-14	2–5
10	D	PH 12-15, PH 12-5	2–5
11	B	PH 12-12, PH 12-6	2–5
12	C	PH 12-12	2–5
13	A	PH 12-12, PH 12-6	2–5
14	B	PH 12-15	2–5
15	A	PH 12-15	2–5
16	D	PH 12-13	3–4
17	A	PH 12-14, PH 12-5	3–5
18	B	PH 12-13, PH 12-4	3–6
19	C	PH 12-12, PH 12-6	3–6
20	C	PH 12-12, PH 12-6	3–6

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Section II

80 marks

Question 21 (6 marks)

Question 21 (a) (1 mark)

Outcomes Assessed: PH 12-13

Targeted Performance Bands: 2–3

Criteria	Marks
• Correctly provides the correct direction of magnetic field	1

Sample Answer:

The magnetic field is directed into the page between the insulated wires.

Question 21 (b) (2 marks)

Outcomes Assessed: PH 12-2, PH 12-13

Targeted Performance Bands: 2–5

Criteria	Marks
• Correctly calculates the distance between the two wires	2
• Uses the formula with an incorrect answer <i>or identifies correct formula</i>	1

Sample Answer:

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

$$0.0138 = \frac{4\pi \times 10^{-7} \cdot 14.4 \times 14.4}{2\pi r}$$

$$0.0138r = 4.1472 \times 10^{-5}$$

$$r = 3.005 \times 10^{-3} \text{ m}$$

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Question 21 (c) (3 marks)**Outcomes Assessed: PH 12-6, PH 12-13****Targeted Performance Bands: 2–5**

Criteria	Marks
• Explains how the flow of current in the wire, production of magnetic fields and the downwards force on the force meter	3
• Describes the force acting on the force meter by referring to magnetic fields	2
• Any relevant information or correct description of magnetic field but incorrect force direction	1

Sample Answer:

When the switch is closed, current is allowed to flow through the parallel conductors. The wire at the top will have a current going to the right and the one on the force meter will have a current going to the left. The current-carrying conductors produce a surrounding magnetic field. The field from the top wire will create a downward force on the bottom wire. (Using right hand push rule)

The lower wire will experience a force downwards, increasing the reading on the force meter.

Question 22 (4 marks)**Question 22 (a) (2 marks)****Outcomes Assessed: PH 12-5, PH 12-12****Targeted Performance Bands: 2–5**

Criteria	Marks
• Correctly calculates the altitude of geostationary orbit showing all working	2
• Correctly calculates the radius of geostationary orbit	1

Sample Answer:

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$\frac{r^3}{86400^2} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{4\pi^2}$$

$$r = \sqrt[3]{\frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 86400^2}{4\pi^2}}$$

$$r = 422297523.87 \text{ m}$$

$$\text{Altitude} = 42.23 \times 10^6 - 6.371 \times 10^6 = 3.59 \times 10^7 \text{ m}$$

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Question 22 (b) (2 marks)**Outcomes Assessed: PH 12-12****Targeted Performance Bands: 2–5**

Criteria	Marks
• Describes the features of a geostationary orbit and justifies the need for a communications satellite to be in geostationary orbit	2
• Describes some features of a geostationary orbit	1

Sample Answer:

A geostationary orbit has the same period of rotation of Earth, and so it keeps a fixed above a specific location on Earth. Its large distance from Earth provides broad coverage and uninterrupted communication, making it ideal for satellite communications.

Question 23 (4 marks)**Question 23 (a) (1 mark)****Outcomes Assessed: PH 12-5, PH 12-13****Targeted Performance Bands: 2–5**

Criteria	Marks
• Calculates the electric field strength with correct units	1

Sample Answer:

$$E = \frac{V}{d}$$

$$E = \frac{10}{0.04}$$

$$E = 250 \text{ Vm}^{-1} \text{ or } \text{NC}^{-1}$$

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Question 23 (b) (3 marks)**Outcomes Assessed: PH 12-6, PH 12-13****Targeted Performance Bands: 2-5**

Criteria	Marks
• Correctly calculates the final velocity of the electron	3
• Correctly calculates work on electron or calculates acceleration of electron	2
• Attempts a calculation using work or electric field strength	1

Sample Answer:

The electron is being accelerated across the entire electric field so,

$$W = qV$$

$$W = 1.602 \times 10^{-19} \times 10$$

$$W = 1.602 \times 10^{-18} \text{ J}$$

Work equals the change in kinetic energy,

$$W = \Delta \frac{1}{2} mv^2$$

$$\frac{1}{2} m(v^2 - u^2) = qV$$

$$v = \sqrt{\frac{2qV}{m} + u^2}$$

$$v = \sqrt{\frac{2 \times 1.602 \times 10^{-19} \times 10}{9.109 \times 10^{-31}} + (3.6 \times 10^5)^2}$$

$$v = 1.9 \times 10^6 \text{ ms}^{-1}$$

Answers could include using:

$$a = \frac{qE}{m} \text{ and } v^2 = u^2 + 2as$$

$$F = qE = ma$$

$$a = \frac{qE}{m} = \frac{1.602 \times 10^{-19} \times 250}{9.109 \times 10^{-31}} = 4.4 \times 10^{13} \text{ ms}^{-2}$$

$$v^2 = u^2 + 2as$$

$$v = \sqrt{(3.6 \times 10^5)^2 + 2(4.4 \times 10^{13})(0.04)} = 1.91 \times 10^6 \text{ ms}^{-1}$$

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Question 24 (8 marks)**Question 24 (a) (4 marks)****Outcomes Assessed: PH 12-13****Targeted Performance Bands: 2–5**

Criteria	Marks
• Explains how an iron core will improve flux linkage, the laminations will reduce eddy current size and relates to heat loss in the transformer	4
• Explains how an iron core will improve flux linkage, the laminations will reduce eddy current size	3
• Describes improved flux linkage and eddy currents OR explains one of them	2
• Some relevant information	1

Sample Answer:

Efficiency measures the useful energy output from the secondary coil as a fraction of the primary input. In Diagram A, the magnetic field from the primary coil spreads out, resulting in poor flux linkage with the secondary coil. A laminated iron core in Diagram B improves flux linkage by directing more of the primary coil's changing magnetic field through the secondary coil, enhancing electromagnetic induction. The laminations reduce eddy currents, minimizing heat loss from resistive heating. These factors increase the secondary coil's output energy and improve transformer efficiency.

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Question 24 (b) (4 marks)**Outcomes Assessed: PH 12-13****Targeted Performance Bands: 2–5**

Criteria	Marks
Explains the decrease in current as rotational speed increases, by referring to induced emf, Lenz's Law and the conservation of energy	4
Explains some aspects of the decrease in current as rotational speed increases	3
Describes how back emf is induced in DC motors	2
Any relevant information	1

Sample Answer:

The graph shows that current flowing through the DC motor decreases as the motor speed increases and the current reaches its lowest value when the motor speed is at its fastest.

As the rotor windings of a DC motor experience a change in magnetic flux during rotation an emf is induced in them. This emf will be in the opposite direction to the supply current and will result in a decrease in the magnitude of the current in the coil. Lenz's Law $\varepsilon = -N \frac{\Delta\Phi}{\Delta t}$ shows that the faster the rotation, the larger the induced emf. The current in the coil therefore $I = \frac{\varepsilon_{\text{supply}} - \varepsilon_{\text{back}}}{R}$. As the motor spins faster the back emf increases and current, consequently, decreases.

This is consistent with the conservation of energy since when current decreases there is less energy transformed per second ($P = VI$). When the current reaches a minimum there will be the least energy transformed per second; just sufficient energy to overcome friction and move the load but not increase the kinetic energy of the system.

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Question 25 (4 marks)**Outcomes Assessed: PH 12-2, PH 12-14****Targeted Performance Bands: 3–5**

Criteria	Marks
• Explanation of TWO ways the apparatus can be changed, supported by an equation	4
• Description of TWO ways the apparatus can be changed or explanation of ONE way, supported by an equation	3
• Description of one way the apparatus can be changed	2
• Any relevant information	1

Sample Answer:

The distance between the light and dark fringes of Young's Double Slit experiment is given by $y = \frac{\lambda L}{d}$ where y is the distance between fringes, λ is the wavelength of light, L is the distance from slits to the screen and d is the slit separation. As wavelength cannot be changed, INCREASING the distance to the screen from the black card and/or DECREASING the slit separation on the card will result in a larger distance between light and dark fringes.

Question 26 (5 marks)**Question 26 (a) (2 marks)****Outcomes Assessed: PH 12-5, PH 12-14****Targeted Performance Bands: 2–5**

Criteria	Marks
• Correctly identifies TWO elements	2
• Correctly identifies ONE element	1

Sample Answer:

Elements Z and A.

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Question 26 (b) (3 marks)**Outcomes Assessed: PH 12-14****Targeted Performance Bands: 2–5**

Criteria	Marks
• Correctly identifies direction of motion and explains how the doppler shift of the spectral lines can determine relative speed	3
• Correctly identifies direction of motion and describes doppler shift	2
• Correctly identifies direction OR identifies doppler shift	1

Sample Answer:

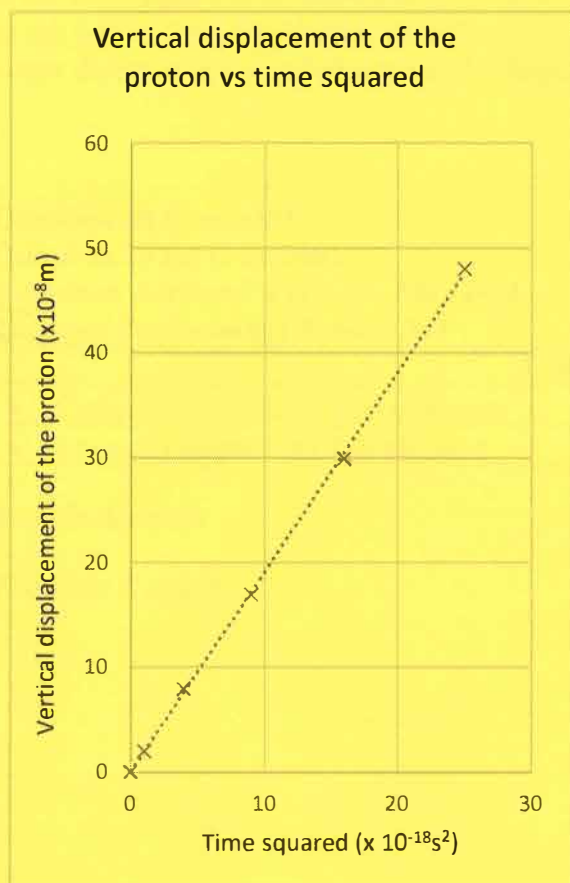
The spectral lines of Yopitrix are identical to those of the element except they have shifted approximately 300nm longer. Due to the Doppler shift of light, it can be assumed that Yopitrix is moving away from Earth. As an object moves away from an observer, the wavelengths of light it emits are stretched.

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Question 27 (6 marks)**Question 27 (a) (4 marks)****Outcomes Assessed: PH 12-4, PH 12-13****Targeted Performance Bands: 2-5**

Criteria	Marks
- Correctly chooses the axes as s versus t^2 - Correctly formats the scale for axes and plots points - Correctly draws a line of best fit - Calculates acceleration using line of best fit	4
Any THREE points from above	3
Any TWO points from above	2
Attempts to plot the data	1

Sample Answer:

$$s_y = u_y t + \frac{1}{2} a_y t^2, \quad u_y = 0$$

$$\frac{a}{2} = \frac{s}{t^2} = \frac{\text{rise}}{\text{run}}$$

$$\frac{\text{rise}}{\text{run}} = \frac{48 \times 10^{-8}}{25 \times 10^{-18}}$$

$$a = \frac{2 \times 48 \times 10^{-8}}{25 \times 10^{-18}}$$

$$a = 3.8 \times 10^{10} \text{ ms}^{-2}$$

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Question 27 (b) (2 marks)**Outcomes Assessed: PH 12-5, PH 12-13****Targeted Performance Bands: 2–5**

Criteria	Marks
• Correctly calculates the time taken using acceleration	2
• Incorrect substitution into correct formula for calculating time OR • Calculates time taken using carried error	1

Sample Answer:

$$s_y = \frac{1}{2} a_y t^2$$

$$15 \times 10^{-3} = \frac{1}{2} (3.84 \times 10^{10}) t^2$$

$$t = 8.8 \times 10^{-7} \text{ s}$$

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Question 28 (5 marks)**Question 28 (a) (2 marks)****Outcomes Assessed: PH 12-5, PH 12-12****Targeted Performance Bands: 2–5**

Criteria	Marks
• Correctly calculates the speed of the bob	2
• Attempts some relevant calculations	1

Sample Answer:

$$\omega = \frac{\Delta\theta}{\Delta t} \text{ rad. s}^{-1}$$

Time for one circumference

$$T = \frac{\Delta\theta}{\omega}$$

$$T = \frac{2\pi}{20}$$

Linear speed

$$v = \frac{2\pi r}{T}$$

$$v = \frac{2\pi \times 0.3}{\frac{2\pi}{20}}$$

$$= 6 \text{ ms}^{-1}$$

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Question 28 (b) (3 marks)**Outcomes Assessed: PH 12-6, PH 12-12****Targeted Performance Bands: 2–5**

Criteria	Marks
• Uses centripetal force and weight force to correctly calculate the angle	3
• Correctly calculates angle but does not show forces acting OR correctly shows forces acting but incorrectly calculates angle	2
• Attempts relevant calculations or forces acting	1

Sample Answer:

$$F_T \cos \theta = mg$$

$$F_T \sin \theta = \frac{mv^2}{r}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1} = \frac{6^2}{0.3 \times 9.8} = 85.3^\circ$$

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Question 29 (4 marks)**Question 29 (a) (1 mark)***Outcomes Assessed: PH 12-15**Targeted Performance Bands: 2–5*

Criteria	Mark
Student correctly identifies the lower region of the Main Sequence	1

Sample Answer:

A mark is placed on the lower half of the shaded main sequence region

Question 29 (b) (3 marks)*Outcomes Assessed: PH 12-15**Targeted Performance Bands: 2–5*

Criteria	Marks
- Explains how the difference in mass of the young stars caused different nucleosynthesis in the core - Explains that supergiants will produce heavier elements in its lifetime, leading to a supernova/black hole/neutron star and white dwarves produces fewer elements - Outlines the evolution from main sequence for both stars	3
- Describes some differences in the nucleosynthesis of the stars - Describes some stages of evolution during the stars' lifetimes	2
- Describes some differences in the nucleosynthesis of the stars OR - Describes some stages of evolution during the stars' lifetimes	1

Sample Answer:

A supergiant undergoes multiple fusion stages, starting with hydrogen fusion via the CNO cycle, followed by helium fusion and the formation of heavier elements up to iron. This is possible due to the immense pressure due to the large mass of the star. Once fusion ceases, the core collapses, leading to a supernova and forming either a neutron star or black hole. In contrast, a white dwarf originates from a low-mass star that fuses hydrogen via the proton-proton chain. After helium fusion in its red giant phase produces carbon and oxygen, fusion stops, and the star sheds its outer layers as a planetary nebula, leaving behind a cooling white dwarf.

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Question 30 (5 marks)**Question 30 (a) (2 mark)****Outcomes Assessed: PH12-5, PH 12-15****Targeted Performance Bands: 2–5**

Criteria	Marks
• Uses a correct procedure to show that one decay produces 1.3041 MeV	2
• Provides some relevant information	1

Sample Answer:

$$\text{Mass defect} = 98.9063 + 0.00054 - 98.9077$$

$$\text{Mass defect} = 0.00086u$$

$$\text{Energy} = 0.00086 \times 931.5$$

$$\text{Energy} = 0.80109 \text{ MeV}$$

Question 30 (b) (3 marks)**Outcomes Assessed: PH 12-6; PH 12-15****Targeted Performance Bands: 3–6**

Criteria	Marks
• Correctly calculates the percentage of molybdenum-99 atoms that have decayed	3
• Shows some working to determine the percentage decay	2
• Provides some relevant information	1

Sample Answer:

$$\lambda = \frac{\ln 2}{\frac{t_1}{2}}$$

$$\lambda = \frac{\ln 2}{65.94}$$

$$\lambda = 0.0105 \text{ hr}^{-1}$$

$$N_t = N_0 e^{-\lambda t}$$

$$\frac{N_t}{N_0} = e^{-(0.0105 \times 5)}$$

$$\frac{N_t}{N_0} = 0.9488$$

$$(1 - 0.9488) \times 100\% = 5.12\% \text{ has decayed.}$$

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Question 31 (4 marks)**Question 31 (a) (2 marks)****Outcomes Assessed: PH 12-5, PH 12-14;****Targeted Performance Bands: 2–5**

Criteria	Marks
• Correctly calculates the momentum of the electron	2
• Provides some relevant information	1

Sample Answer:

$$p_v = \frac{m_0 v}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$p_v = \frac{(9.109 \times 10^{-31}) \times (0.4) \times 3 \times 10^8}{\sqrt{\left(1 - \frac{0.4^2 \times c^2}{c^2}\right)}}$$

$$p_v = 1.193 \times 10^{-22} \text{ kgms}^{-1}$$

Question 31 (b) (2 mark)**Outcomes Assessed: PH 12-14****Targeted Performance Bands: 3–6**

Criteria	Marks
• Explains why the momentum of an electron travelling at 0.8c is more than double an electron at 0.4c	2
• Provides some relevant information	1

Sample Answer:

The reason the momentum of an electron traveling at 0.8c is more than double that of an electron traveling at 0.4c is because relativistic momentum increases more sharply as the velocity approaches the speed of light.

While both electrons have the same rest mass, the Lorentz factor increases significantly as the velocity increases. Specifically, the Lorentz factor at 0.8c is 1.667, whereas at 0.4c it is 1.095.

Thus, the electron traveling at 0.8c has more than double the momentum of the electron traveling at 0.4c because the Lorentz factor grows non-linearly as velocity increases.

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Question 32 (5 marks)**Question 32 (a) (3 marks)****Outcomes Assessed: PH 12-6, PH 12-14****Targeted Performance Bands: 2–5**

Criteria	Marks
- Accounts for the production of both curves via black body radiation - Accounts for the different peak wavelength of each curve - Accounts the dips in curve B due to absorption of light in the outer atmosphere of the star	3
Any TWO points above	2
Provides any relevant information	1

Sample Answer:

Curves A and B represent black body radiation from hot objects, which emit more radiation at longer wavelengths and less at shorter ones. Curve A has a longer peak wavelength, indicating a lower temperature per Wien's Displacement Law. The dips in curve B result from the star's outer atmosphere absorbing specific wavelengths, reducing their observed intensity.

Question 32 (b) (2 marks)**Outcomes Assessed: PH 12-5, PH 12-15****Targeted Performance Bands: 2–5**

Criteria	Marks
Correctly calculates the surface temperature of a star	2
Attempts a calculation using Wien's Displacement Law	1

Sample Answer:

$$\lambda_{\max} = 410 \times 10^{-9} \text{ m (Can accept values from 390 – 420nm)}$$

$$\lambda_{\max} = \frac{b}{t}$$

$$t = \frac{b}{\lambda_{\max}}$$

$$t = \frac{2.898 \times 10^{-3}}{410 \times 10^{-9}}$$

$$t = 7068 \text{ K}$$

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Question 33 (4 marks)**Outcomes Assessed: PH 12-7, PH 12-15****Targeted Performance Bands: 3–6**

Criteria	Marks
• Outlines the role of energy-mass equivalence, and ALL fundamental particles and forces involved in the production of a Helium atom	4
• Outlines the role of energy-mass equivalence, and MOST fundamental particles and forces involved in the production of a Helium atom	3
• Outlines the particles and forces that make up a Helium atom	2
• Provides some relevant information	1

Sample Answer:

Matter can spontaneously form when energy converts into mass, as described by $E = mc^2$, since matter and energy are interchangeable. The first matter consisted of fundamental particles—quarks and leptons.

A helium-4 atom has a nucleus of two protons and two neutrons, surrounded by two electrons. These atoms formed through nucleosynthesis, where protons and neutrons combined into helium-4 nuclei.

Each proton consists of two up quarks and one down quark, while each neutron has two down quarks and one up quark. The strong nuclear force (SNF) binds quarks within protons and neutrons and also holds the nucleus together, counteracting the electromagnetic force (EMF) that repels protons.

The EMF also governs the interaction between the nucleus and electrons, influencing their motion as described in the Schrödinger model. Gravity, being negligible compared to other forces, has no significant role in the helium atom.

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Question 34 (8 marks)**Outcomes Assessed:** PH 12-12, PH 12-14, PH 12-15, PH 12-6; PH12-7**Targeted Performance Bands:** 2–6

Criteria	Marks
• Outlines the operation of particle accelerators such as the LHC to collide protons. • Explains that extra mass can arise from the relativistic momentum of fast-moving protons • Explains the relativistic momentum and kinetic energy increase of each proton will be transformed into the masses of the larger quarks via mass-energy equivalence • Correctly calculates the minimum energy requirements of the collisions using mass-energy equations • Correctly calculates the minimum velocity of the protons required to produce top and bottom quarks	7–8
• Any 4 of the Points above or makes an error in calculation	6–7
• Describes the function of particle accelerators • Outlines mass-energy equivalence • Attempts a calculation • Describes momentum dilation	4–5
• Any THREE points above	2–3
• Any relevant information	1

Sample Answer:

Particle accelerators, such as the Large Hadron Collider (LHC), accelerate protons to nearly the speed of light using powerful electric and magnetic fields. They are then collided in opposite direction, creating new particles. As the protons are travelling close to the speed of light, their momentum is significantly dilated, contributing to increased kinetic energy. In a collision, the energy from the high-speed protons is converted into mass, as stated in Einstein's equation $E = mc^2$. To produce heavy quarks like the top and bottom quarks, the energy involved in the collision must be sufficient to create these quarks' mass.

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$$E_{total} = E_{top} + E_{bottom}$$

$$E_{total} = 173 \text{ GeV} + 4.18 \text{ GeV}$$

$$E_{total} = 177.18 \text{ GeV}$$

Mass Equivalent:

$$\frac{177.18 \times 10^3}{931.5} \times 1.661 \times 10^{-27}$$

$$m_v = 3.16 \times 10^{-25} \text{ kg}$$

Mass of two protons:

$$m_0 = 2 \times 1.673 \times 10^{-27}$$

$$m_0 = 3.346 \times 10^{-27} \text{ kg}$$

Using momentum dilation:

$$p_v = \frac{p_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\sqrt{1 - \frac{v^2}{c^2}} = \frac{m_0 v}{m_v v}$$

$$v = \sqrt{c^2 \times \left(1 - \frac{m_0^2}{m_v^2}\right)}$$

$$v = \sqrt{(3 \times 10^8)^2 \times 1 - \frac{(3.346 \times 10^{-27})^2}{(3.16 \times 10^{-25})^2}}$$

$$v = 0.999944c$$

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Question 35 (8 marks)**Outcomes Assessed: PH 12-12, PH 12-13, PH 12-14, PH 12-15, PH12-7****Targeted Performance Bands: 2–6**

Criteria	Marks
- Extensively outlines a model which was created due to the results of an experiment - Extensively outlines a model which was validated due to the results of an experiment - Extensively outlines a model which was abandoned due to the results of an experiment - Justifies the importance of experimental proof to change the nature of Physics	7–8
- Outlines a model which was created due to the results of an experiment - Outlines a model which was validated due to the results of an experiment - Outlines a model which was abandoned due to the results of an experiment - Justifies the importance of experimental proof to change the nature of Physics	5–6
Recalls TWO examples of models and experiments with a justification	3–4
ONE example of a model and experiment is given	2
Some relevant information	1

Sample Answer:

Answers could include:

Creation of a new Model

- Geiger-Marsden's Gold Foil Experiment
 - After firing alpha particles at gold foil, it was found that most particles went straight through the gold with some deflecting at large angles. This meant that atoms of matter were concentrated in a dense nucleus, surrounded by empty space. This began the nuclear model of the atom as proposed by Rutherford.
- Chadwick's discovery of neutrons
 - Although Rutherford had theorised that the nucleus of an atom contained particles with similar mass to protons but without charge (a combination of a proton and electron that could not be explained), Chadwick used conservation of momentum to determine the mass of a neutron and validated the nuclear model of the atom. He concluded these particles were new, unique particles of the nucleus of an atom.

Validation of a model

- Millikan's Oil Drop Experiment
 - Millikan was able to measure the fundamental unit of charge – the charge of a single electron. Validated the idea that electrons are unique, subatomic particles and theorised by JJ Thomson after his charge to mass ratio experiments.

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- de Broglie's matter waves
 - Davisson and Germer detected the wave interference pattern of electrons as they reflected off a Nickel target. This validated de Broglie's model of wave-particle duality and matter waves.
- Hafele-Keating Atomic Clocks
 - After flying very sensitive, synchronised atomic clocks with varying relative velocities, they returned different times when brought back together. This validated time dilation, as proposed in Einstein's Special Theory of Relativity.
- Hubble and the expanding universe
 - Hubble measured the recession of other galaxies using doppler shift and found they were all (except one) moving rapidly away from Earth. He concluded the universe is expanding. This was first theorised by Friedmann so validated this model.

Abandoned Models

- Quantised energy due to the Ultraviolet Catastrophe in black body radiation.
 - Energy was required to be quantised to fit the experimental evidence as proposed with Wien's law and black body radiation. $E = hf$ was devised which related energy and the frequency of an oscillator. This meant that the energy emitted and absorbed by electrons in the atom could not simply come in any amount, as proposed by classical physics.
- Photoelectric experiments and quantized light.
 - showed the kinetic energy of photoelectrons was proportional to the frequency of light and that each metal had a particular threshold frequency which needed to be overcome. This contradicted the classical wave nature of light. Einstein suggested that light was quantized and had a wave-particle duality. As it interacted with matter, it behaved as a particle.